

MSDM5004 Spring 2026

Exercise 4

Mar 9

We solve the equation $u_t - 2u_x = 0$ using the upwind scheme. Consider the numerical solution U_j^{n+1} at the grid point (x_j, t_{n+1}) .

Derive the CFL condition based on the domain of dependence of the exact solution and numerical domain of dependence on the time level t_n .

Solution.

The upwind scheme of this equation is

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} - 2 \frac{U_{j+1}^n - U_j^n}{\Delta x} = 0,$$

or

$$U_j^{n+1} = U_j^n + 2\nu (U_{j+1}^n - U_j^n), \quad \nu = \frac{\Delta x}{\Delta t}.$$

The numerical domain of dependence on time level t_n is $x_j \leq x \leq x_{j+1}$.

The characteristic passing through the point (x_j, t_{n+1}) is

$$x + 2t = x_j + 2t_{n+1}.$$

It intersects with time level t_n at

$$x = -2t_n + x_j + 2t_{n+1} = x_j + 2\Delta t,$$

and we have $u(x_j, t_{n+1}) = u(x_j + 2\Delta t, t_n)$ due to the fact that the solution of the PDE is constant along a characteristic. Thus the domain of dependence of the exact solution on time level t_n is $\{x_j + 2\Delta t\}$.

The CFL condition requires that the domain of dependence is inside the numerical domain of dependence, which is

$$x_j \leq x_j + 2\Delta t \leq x_{j+1} = x_j + \Delta x.$$

Thus we have

$$2\Delta t \leq \Delta x,$$

or

$$2\nu \leq 1.$$

This is the CFL condition.

